

Qualification Pack

Biomedical Equipment Maintenance Assistant

QP Code: NIE/ELE/Q0802

Version: 1.0

NSQF Level: 3

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NIE/ELE/Q0802: Biomedical Equipment Maintenance Assistant

Brief Job Description

The course covers tools and techniques for servicing medical equipment, including soldering and test instruments, while exploring electrical concepts like current, voltage, and power, elucidating passive component principles such as resistors, capacitors, and inductors. Detailed study of active components like semiconductors and integrated circuits offers insights into design, testing, and applications in medical instrumentation. Biomedical instrumentation topics encompass the man-instrument system, physiological systems, bio-electric potentials, and microscope usage, alongside thorough examination of ECG machines, pH meters, patient monitors, and diagnostic equipment, including their principles, testing, and calibration. The course culminates with a focus on employability skills to prepare learners for success in the medical equipment and devices industry.

Personal Attributes

A proficient biomedical equipment maintenance assistant should possess sharp observational skills and a solid understanding of electronics fundamentals, including voltage and current waveforms. They must demonstrate practical expertise in using tools such as multimeters, CROs, and soldering on PCBs. Additionally, a strong foundation in biomedical engineering principles and safety protocols is essential for ensuring patient and equipment safety. Effective communication and collaborative abilities are necessary for analyzing and troubleshooting a range of biomedical instruments, from ECG machines to microscopes. Attention to detail and a methodical approach are crucial for performing maintenance tasks with precision and efficiency.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0808: Introduction to Healthcare Delivery Systems and Role of Medical Equipment and Devices](#)
2. [NIE/ELE/N0809: Electronics Fundamentals and Laboratory Techniques](#)
3. [NIE/ELE/N0807: Biomedical Diagnostics: Instrumentation and Analysis](#)
4. [NIE/ELE/N0816: Implementation of Biomedical Equipment project](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
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Sub-Sector	Medical Electronics
Occupation	Electronics Equipment Services
Country	India
NSQF Level	3
Credits	10
Aligned to NCO/ISCO/ISIC Code	3211.0501
Minimum Educational Qualification & Experience	10th grade pass OR Grade 8 pass with 2 years of (NTC/ NAC) after 8th with NA of experience OR Not Applicable (▲Previous relevant qualification of NSQF Level 2) with 1 Year of experience OR Not Applicable (▲Previous relevant qualification of NSQF Level 2.5) with 6 Months of experience OR 8th grade pass with 2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQF Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-03-EH-01966-2024-V1-NIELIT
NQR Version	1

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NIE/ELE/N0808: Introduction to Healthcare Delivery Systems and Role of Medical Equipment and Devices

Description

This course delves into healthcare delivery systems and the indispensable role of medical equipment and devices. It explores various models, stakeholders, and regulatory factors shaping healthcare services globally. Students will examine the lifecycle of medical equipment, from design to disposal, and its integration into healthcare systems. By course completion, students will be equipped to navigate complexities and contribute to advancements in healthcare delivery practices.

Scope

The scope covers the following :

- The course encompasses developing students' abilities to analyze and understand the fundamental structure and operations of healthcare delivery systems, with a specific focus on India. Students will learn to compile comprehensive reports summarizing their observations regarding the basic structure and functions of India's healthcare delivery system. Additionally, they will gain proficiency in identifying basic medical devices and equipment crucial for healthcare delivery.

Elements and Performance Criteria

Healthcare Delivery System and Medical Equipment Overview

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the basic structure and function of the healthcare delivery system in India and Identify the basic medical equipment and devices.

Understanding Healthcare Delivery Systems and Services

To be competent, the user/individual on the job must be able to:

- PC2.** Distinguish between private, public, and non-profit healthcare delivery systems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students will develop skills to analyze and comprehend the fundamental structure and operations of healthcare delivery systems, particularly focusing on India.
- KU2.** Students will learn to compile comprehensive reports summarizing their observations regarding the basic structure and functions of India's healthcare delivery system.
- KU3.** Students will gain expertise in identifying basic medical devices and equipment essential for effective healthcare delivery.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Students will develop the ability to critically analyze and synthesize information about the basic structure and function of India's healthcare delivery system, enabling them to prepare comprehensive reports summarizing their observations effectively.
- GS2.** Students will develop proficiency in identifying essential medical devices and equipment crucial for healthcare delivery, showcasing competence in recognizing and comprehending the tools utilized in clinical settings.
- GS3.** Through report preparation and device identification exercises, students will enhance their communication skills, both written and verbal, enabling them to effectively convey their findings and insights to various stakeholders within the healthcare field.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Healthcare Delivery System and Medical Equipment Overview</i>	20	-	-	-
PC1. Describe the basic structure and function of the healthcare delivery system in India and Identify the basic medical equipment and devices.	-	-	-	-
<i>Understanding Healthcare Delivery Systems and Services</i>	-	-	-	-
PC2. Distinguish between private, public, and non-profit healthcare delivery systems.	-	-	-	-
NOS Total	20	-	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0808
NOS Name	Introduction to Healthcare Delivery Systems and Role of Medical Equipment and Devices
Sector	Electronics
Sub-Sector	Medical Electronics
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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NIE/ELE/N0809: Electronics Fundamentals and Laboratory Techniques

Description

Electronics fundamentals and laboratory techniques encompass the basic principles, theories, and practical skills essential for understanding and working with electronic circuits and devices.

Scope

The scope covers the following :

- This course providing students with a strong foundation in electronics theory and practical skills. students will have a thorough understanding of voltage and current waveforms, along with proficiency in using essential tools such as multimeters and cathode ray oscilloscopes (CRO). They will be able to identify and calculate values for passive components, comprehend magnetism and transformer principles, and effectively utilize various active components in electronic circuits. Additionally, students will acquire hands-on experience in soldering and de-soldering techniques on printed circuit boards (PCBs). This comprehensive scope prepares students for successful careers in electronics engineering, maintenance, and technical support roles, equipped with the knowledge and skills necessary to troubleshoot, analyze, and repair electronic systems and devices effectively.

Elements and Performance Criteria

Introduction to Electrical Fundamentals and Maintenance Tools

To be competent, the user/individual on the job must be able to:

- PC1.** Introduction to the fundamental concepts of electricity and essential tools used in servicing and maintenance tasks
- PC2.** Understand voltage, current, power, and AC waveform principles, along with the tools required for effective maintenance operations

Practical Applications of Test Equipment and Resistance Fundamentals

To be competent, the user/individual on the job must be able to:

- PC3.** Usage of essential test equipment such as multimeters, cathode ray oscilloscopes (CRO), signal generators, and LCR meters.
- PC4.** Understand fundamental concepts related to resistance, including value calculation, types, working principles, and testing methodologies.

Capacitors, Inductors, and Transformers: Theory and Practical Applications

To be competent, the user/individual on the job must be able to:

- PC5.** Knowledge of capacitors, inductors and introduction to transformers. student will delve into the theory, types, value calculation methods, working principles, and testing procedures of capacitors and inductors.
- PC6.** Learn about magnetic materials, self and mutual inductance, and the operation of transformers, including step-up and step-down configurations.

Semiconductor Theory, Active Components, ICs, and Soldering Techniques

To be competent, the user/individual on the job must be able to:

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PC7. Learn semiconductor theory, active electronic components, integrated circuits (ICs), soldering techniques and soldering and De-soldering Techniques.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Students will develop a comprehensive understanding of voltage and current waveforms, laying the groundwork for analyzing electrical signals in circuits.
- KU2.** Students will master the use of multimeters and cathode ray oscilloscopes (CRO), enabling accurate measurement and analysis of electronic signals.
- KU3.** Students will learn to identify and calculate values for passive components like resistors, capacitors, and inductors, enhancing their circuit analysis and design skills.
- KU4.** Students will grasp the principles of magnetism and transformer operation, enabling them to comprehend the functionality and applications of transformers in electronic systems.
- KU5.** Students will gain practical experience in utilizing active components such as diodes, transistors, and integrated circuits (ICs) effectively in electronic circuit design and troubleshooting, preparing them for careers in electronics engineering and maintenance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop strong analytical skills through understanding voltage and current waveforms and applying them to analyze electrical signals in circuits, essential for problem-solving in electronics engineering and maintenance roles.
- GS2.** Proficiency with multimeters and cathode ray oscilloscopes (CRO) empowers students to skillfully measure, analyze, and interpret electronic signals, enhancing their diagnostic capabilities and technical acumen in troubleshooting electronic systems.
- GS3.** Gain hands-on experience in identifying, calculating values for, and utilizing passive components and active components in electronic circuit design and troubleshooting scenarios, fostering practical skills

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Electrical Fundamentals and Maintenance Tools</i>	80	40	-	-
PC1. Introduction to the fundamental concepts of electricity and essential tools used in servicing and maintenance tasks	-	-	-	-
PC2. Understand voltage, current, power, and AC waveform principles, along with the tools required for effective maintenance operations	-	-	-	-
<i>Practical Applications of Test Equipment and Resistance Fundamentals</i>	-	-	-	-
PC3. Usage of essential test equipment such as multimeters, cathode ray oscilloscopes (CRO), signal generators, and LCR meters.	-	-	-	-
PC4. Understand fundamental concepts related to resistance, including value calculation, types, working principles, and testing methodologies.	-	-	-	-
<i>Capacitors, Inductors, and Transformers: Theory and Practical Applications</i>	-	-	-	-
PC5. Knowledge of capacitors, inductors and introduction to transformers. student will delve into the theory, types, value calculation methods, working principles, and testing procedures of capacitors and inductors.	-	-	-	-
PC6. Learn about magnetic materials, self and mutual inductance, and the operation of transformers, including step-up and step-down configurations.	-	-	-	-
<i>Semiconductor Theory, Active Components, ICs, and Soldering Techniques</i>	-	-	-	-
PC7. Learn semiconductor theory, active electronic components, integrated circuits (ICs), soldering techniques and soldering and De-soldering Techniques.	-	-	-	-
NOS Total	80	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0809
NOS Name	Electronics Fundamentals and Laboratory Techniques
Sector	Electronics
Sub-Sector	Medical Electronics
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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NIE/ELE/N0807: Biomedical Diagnostics: Instrumentation and Analysis

Description

It encompasses a comprehensive curriculum focusing on the development and utilization of biomedical instrumentation to understand human physiological systems.

Scope

The scope covers the following :

- Biomedical Instrumentation Development
- Exploration of Biomedical Techniques
- Integration of Computing in Biomedical Domain
- Safety Protocols and Instrument Operation
- Maintenance and Troubleshooting

Elements and Performance Criteria

Biomedical Instrumentation Development

To be competent, the user/individual on the job must be able to:

- PC1.** Student demonstrates an understanding of the fundamental principles underlying the design and development of biomedical instrumentation, including signal processing, sensor technologies, and system integration.
- PC2.** Student applies theoretical knowledge to practical scenarios by designing and prototyping biomedical instruments, incorporating appropriate sensors, amplifiers, and signal conditioning circuits to achieve accurate and reliable measurements.
- PC3.** Student evaluates the performance of biomedical instruments through experimental testing, identifies areas for improvement, and optimizes design parameters to enhance accuracy, sensitivity, and usability.

Exploration of Biomedical Techniques

To be competent, the user/individual on the job must be able to:

- PC4.** Student demonstrates proficiency in utilizing a variety of biomedical techniques such as bio-electric potentials, cardiovascular analysis, microscopic examination, and pH measurement, accurately interpreting experimental data and results.
- PC5.** Student designs and executes experiments effectively, selecting appropriate techniques and instrumentation to address specific biomedical research questions or diagnostic challenges, while considering factors such as sample preparation, data acquisition, and analysis.
- PC6.** Student analyzes experimental data using statistical methods, graphical visualization, and qualitative analysis techniques, interpreting results to draw meaningful conclusions and make informed decisions in biomedical diagnostics and research.

Integration of Computing in Biomedical Domain

To be competent, the user/individual on the job must be able to:

- PC7.** Student understands the principles of digital computing and microprocessor-based systems, including programming languages, data structures, algorithms, and interfacing techniques relevant to biomedical applications.

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- PC8.** Student integrates computing technologies with biomedical instrumentation effectively, designing software interfaces, developing data acquisition and processing algorithms, and implementing control systems to enhance the functionality and performance of biomedical devices.
- PC9.** Student troubleshoots software and hardware issues in biomedical computing systems, employing systematic debugging techniques, diagnostic tools, and problem-solving strategies to identify and resolve technical problems in a timely manner.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Fundamental Principles of Biomedical Instrumentation:
• Understanding the theoretical foundations and principles governing the design, development, and operation of biomedical instrumentation, including signal processing, sensor technologies, and system integration.
- KU2.** • Biomedical Techniques and Methods:
• Knowledge of a diverse range of biomedical techniques and methods such as bio-electric potentials, cardiovascular analysis, microscopic examination, pH measurement, patient monitoring, and ECG analysis, along with their underlying principles and applications in biomedical diagnostics and research.
- KU3.** • Integration of Computing Technologies:
• Understanding the integration of digital computing and microprocessor-based systems in the biomedical domain, including programming languages, data structures, algorithms, interfacing techniques, and software development principles relevant to biomedical applications.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Technical Proficiency Skills:
• Demonstrating proficiency in designing, developing, and operating biomedical instrumentation, along with utilizing various biomedical techniques and methods effectively to address research questions or diagnostic challenges.
- GS2.** • Analytical Skills:
• Applying analytical skills to interpret experimental data, evaluate the performance of biomedical instruments, and make informed decisions in biomedical diagnostics, research, and development.
- GS3.** • Problem-solving and Innovation Skills:
• Utilizing problem-solving skills to troubleshoot technical issues, optimize instrument performance, and innovate novel solutions in biomedical instrumentation and analysis, contributing to advancements in healthcare technology and patient care.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Biomedical Instrumentation Development</i>	100	50	-	-
PC1. Student demonstrates an understanding of the fundamental principles underlying the design and development of biomedical instrumentation, including signal processing, sensor technologies, and system integration.	-	-	-	-
PC2. Student applies theoretical knowledge to practical scenarios by designing and prototyping biomedical instruments, incorporating appropriate sensors, amplifiers, and signal conditioning circuits to achieve accurate and reliable measurements.	-	-	-	-
PC3. Student evaluates the performance of biomedical instruments through experimental testing, identifies areas for improvement, and optimizes design parameters to enhance accuracy, sensitivity, and usability.	-	-	-	-
<i>Exploration of Biomedical Techniques</i>	-	-	-	-
PC4. Student demonstrates proficiency in utilizing a variety of biomedical techniques such as bio-electric potentials, cardiovascular analysis, microscopic examination, and pH measurement, accurately interpreting experimental data and results.	-	-	-	-
PC5. Student designs and executes experiments effectively, selecting appropriate techniques and instrumentation to address specific biomedical research questions or diagnostic challenges, while considering factors such as sample preparation, data acquisition, and analysis.	-	-	-	-
PC6. Student analyzes experimental data using statistical methods, graphical visualization, and qualitative analysis techniques, interpreting results to draw meaningful conclusions and make informed decisions in biomedical diagnostics and research.	-	-	-	-
<i>Integration of Computing in Biomedical Domain</i>	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC7. Student understands the principles of digital computing and microprocessor-based systems, including programming languages, data structures, algorithms, and interfacing techniques relevant to biomedical applications.	-	-	-	-
PC8. Student integrates computing technologies with biomedical instrumentation effectively, designing software interfaces, developing data acquisition and processing algorithms, and implementing control systems to enhance the functionality and performance of biomedical devices.	-	-	-	-
PC9. Student troubleshoots software and hardware issues in biomedical computing systems, employing systematic debugging techniques, diagnostic tools, and problem-solving strategies to identify and resolve technical problems in a timely manner.	-	-	-	-
NOS Total	100	50	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0807
NOS Name	Biomedical Diagnostics: Instrumentation and Analysis
Sector	Electronics
Sub-Sector	Medical Electronics
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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NIE/ELE/N0816: Implementation of Biomedical Equipment project

Description

The course provides training in servicing and maintaining medical equipment, covering essential skills like soldering, de-soldering, and using test instruments. It teaches basic electrical concepts and the function of both passive and active components. It also explores biomedical instrumentation, including physiological systems, bio-electric potentials, and medical devices such as ECG machines, pH meters, and patient monitors. The course ends with employability skills to prepare students for careers in the medical device industry.

Scope

The scope covers the following :

- The course covers key National Occupational Standards (NOS) for Biomedical Equipment Maintenance Assistants. It includes an introduction to healthcare systems and the role of medical devices, along with fundamentals of electronics and laboratory techniques such as soldering, measuring, and testing components. Learners gain hands-on experience with biomedical diagnostic equipment like ECG machines, pH meters, and microscopes, focusing on maintenance and safety. The course also emphasizes employability skills including communication, digital literacy, and workplace readiness. These NOS units collectively prepare learners for entry-level roles in the medical device industry.

Elements and Performance Criteria

Perform basic maintenance and testing of biomedical equipment and devices.

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and explain the working principles of electronic components and diagnostic medical equipment.
- PC2.** Demonstrate the use of tools and instruments for soldering, de-soldering, testing, and calibration.
- PC3.** Follow standard procedures and safety protocols during equipment maintenance and diagnostics.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learners gain knowledge of the Indian healthcare system and the essential role biomedical devices play in diagnostics and patient care.
- KU2.** Covers core electrical principles and the function of passive and active electronic components, along with hands-on use of testing and measuring instruments.
- KU3.** Provides insight into physiological systems, bio-electric signals, and the working of diagnostic tools like ECG machines and pH meters, with emphasis on safety and calibration.

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KU4. Includes practical skills such as soldering, de-soldering, basic maintenance, and fault finding, along with soft skills like communication, digital literacy, and workplace readiness.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to communicate clearly and effectively with team members, supervisors, and healthcare staff.
- GS2.** Proficient in using basic computer applications, internet, and digital tools for documentation and communication.
- GS3.** Demonstrates cooperation, adaptability, and positive interaction in group settings.
- GS4.** Capable of managing tasks efficiently, prioritizing work, and utilizing tools and materials responsibly.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform basic maintenance and testing of biomedical equipment and devices.</i>	-	-	-	30
PC1. Identify and explain the working principles of electronic components and diagnostic medical equipment.	-	-	-	-
PC2. Demonstrate the use of tools and instruments for soldering, de-soldering, testing, and calibration.	-	-	-	-
PC3. Follow standard procedures and safety protocols during equipment maintenance and diagnostics.	-	-	-	-
NOS Total	-	-	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0816
NOS Name	Implementation of Biomedical Equipment project
Sector	Electronics
Sub-Sector	
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the
- defined learning outcomes and assessment criteria.
- The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate
- can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the
- practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is
- required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of

Qualification Pack

NIELIT HQS.

2. The assessor assigns roll number

3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

- Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0808.Introduction to Healthcare Delivery Systems and Role of Medical Equipment and Devices	20	0	0	0	20	6
NIE/ELE/N0809.Electronics Fundamentals and Laboratory Techniques	80	40	0	0	120	32
NIE/ELE/N0807.Biomedical Diagnostics: Instrumentation and Analysis	100	50	0	0	150	40
NIE/ELE/N0816.Implementation of Biomedical Equipment project	0	0	0	30	30	8
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	14
Total	220	120	-	30	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
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Qualification Pack

NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards.
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
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